2025W2 BIOL 300 Midterm (Section 202, AM)

Instructions:

1. Make sure that your copy of the test includes 10 pages, including this one.
2. Please round final answers as specified within the question (if rounding is needed). We will mark final answers with a tolerance for intermediate rounding but aim to retain 4-5 decimal places in intermediate calculations.
3. Some questions have a box for the final answer. Please put the final answer in the box and show your work in the other space provided.
4. We recommend writing your student ID # at the top of each page.
5. Answers can be in ink or pencil.
6. Other than a basic/scientific calculator, no electronic devices are permitted. Please turn off your cell phone and leave it in your bag.
7. Students who engage in dishonest practices will be subject to disciplinary action.
8. By taking this exam, you are declaring that your answers are all your own work.
9. Students taking the exam may not speak or in any other way communicate with other students or people outside the examination room.
10. Students may not expose their written paper to other students. The excuse of accidental exposure, forgetfulness, or ignorance will not be accepted.

Please fill out the box below. By signing, you acknowledge that you have read and fully understand the instructions above.

Student number								
Section								
Signature								
Name								

Student Number:_____

1. Use the sample of measurements below (all in cm) for all questions on this page. Pay attention to the units and remember to convert units if needed (1 cm = 10 mm; 1 m = 100 cm).

   17, 29, 32, 4, 15, 23

   a. (2 points) What is the sample mean, in **cm**?

      mean = (17+29+32+4+15+23)/6 = 20

      Answer: 20

   b. (3 points) What is the sample standard deviation, in **cm**, rounded to four decimal places?

      sd = 10.23719 cm, which rounds to 10.2372

      Answer: 10.2372

   c. (2 points) What is the sample median, in **mm**?

      sorted: 4 15 17 23 29 32
      take mean of 17 & 23 -> 20 cm, which is 200 mm

      Answer: 200

   d. (2 points) What is the standard error of the mean, in **cm**, rounded to four decimal places?

      se = 10.23719 / sqrt(6) = 4.179314, which rounds to 4.1793

      Answer: 4.1793

   e. (2 points) What is the sample variance, in $\mathbf{m^2}$? Round to five decimal places.

      sample variance = 10.23719^2 = 104.8 cm^2
      1cm = 0.01m, so multiply by 0.01^2 a.k.a. 0.0001 -> 0.01048 m^2

      Answer: 0.01048

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2.

a. (2 points) In a sample of 66 island lizards, the mean hindlimb length was 68.0 mm, with a 95% CI of 67.4 to 68.6 mm. Assuming this variable is normally distributed, circle *all* that are true.

   i. The standard error of the mean is approximately 0.3 mm **Yes**
   ii. The standard error of the mean is approximately 0.3 mm^2
   iii. The standard error of the mean is approximately 0.6 mm
   iv. The standard deviation of the data is approximately 0.3 mm
   v. The standard deviation of the data is approximately 0.09 mm^2

b. (3 points) Which among the following are appropriate interpretations of the standard error of the mean? Circle *all* that apply.

   i. The amount by which the sample mean is expected to differ from the population mean in any single random sample.
   ii. The standard deviation of the sampling distribution of the mean. **Yes**
   iii. A measure of the variability of individual observations in the sample.
   iv. A measure of the variability of the true population mean.
   v. To measure it, compute the sample mean and then shout "standard error" really loudly.
   vi. None of the above.

c. (2 points) Say that we choose to compute a 99% CI. To what value of α would that conceptually correspond? Give your answer in decimal form, with no more than 2 decimal places.

   Answer: 0.01

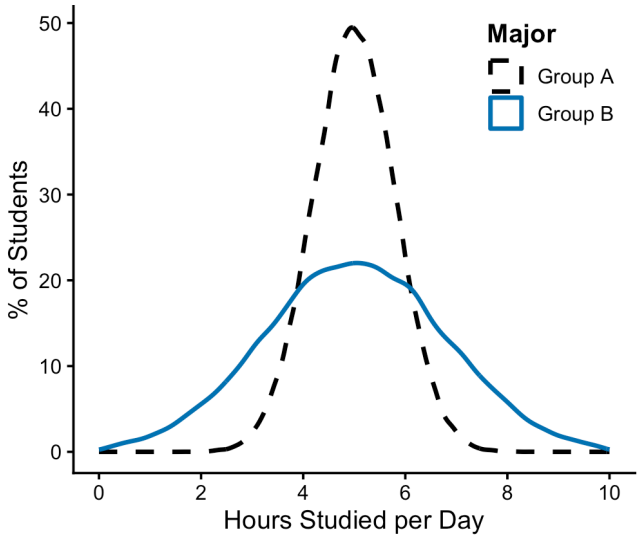
d. (3 points) If we choose to use a smaller value of α (e.g., smaller than 0.05), we risk increasing the probability of making a certain kind of error. Briefly explain what this error is in 1-3 sentences.

   Using a smaller α (e.g., 0.01 instead of 0.05) increases the probability of a **Type II** error. This type of error is one where we fail to reject a null hypothesis, and the null hypothesis was really false. Answer can optionally include: "missing a real effect" or "reduces power"

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3.

The figure below shows distributions of hours studied per day for separate, large samples of students from two different majors (Group A vs. Group B). The sample size of each group was roughly equal. No student is in both groups.



Based on what you can safely conclude from the figure, mark each statement below as True or False in the column on the right (2 points each).

The mean of Group A is greater than the mean of Group B.	**False**
The mode of Group A is greater than the mode of Group B.	**False**
Group B has a larger standard deviation than Group A.	**True**
The population of Group A is larger than the population of Group B.	**False**
A randomly chosen student from Group B is likely to have studied more hours than a randomly chosen student from Group A.	**False**
A student who studied 8 hours per day is more likely to be from Group B than Group A.	**True**

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4. Researchers conducted a randomized, placebo-controlled study to test whether a drug reduces the risk of a new breast cancer event. Participants were randomly assigned to receive either the drug or a placebo and were followed for several years. In the drug treatment group, 14 out of 253 participants experienced a new breast cancer event. In the placebo group, 28 out of 247 experienced a new breast cancer event.

   a. (12 points) Is there sufficient evidence of an association between treatment group and the risk of a new breast cancer event? Conduct an appropriate statistical test. Assume $\alpha = 0.05$. Clearly state null and alternative hypotheses and the name of the test you are using. Give the P-value as precisely as possible given the statistical tables, and interpret your result in words. Assume random sampling and an unbiased study design.

H0: Treatment group and event status are independent (no association).
HA: Treatment group and event status are not independent (association).
(exact wording of hypotheses may vary)

chi-squared contingency analysis (/test)

OBS	event	no event	
drug	14	239	253
placebo	28	219	247
	42	458	500

EXP	event	no event	
drug	21.2520	231.7480	253
placebo	20.7480	226.2520	247
	42	458	500

Chi-squared = 2.4747 + 2.5348 + 0.22693 + 0.23245 = 5.4688
df = 1

Based on table, we can infer P-value is between 0.025 and 0.01

Since P < alpha, we reject the null hypothesis. The evidence suggests an association between treatment group and risk of a new breast cancer event. Specifically, the event rate is lower in the drug group (14/253 ≈ 5.5%) than placebo (28/247 ≈ 11.3%)

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b. (4 points) Calculate the 95% CI confidence interval for the population proportion of drug-treated individuals who would experience a breast cancer event, as described on the previous page. Round each value in your answer to four decimal places.

p' = (14+2) / (253 + 4) = 0.0622568

$$0.0622568 \pm 1.96\sqrt{\frac{0.0622568(1-0.0622568)}{257}}$$

$$0.0622568 \pm 1.96(0.0150719)$$

Lower : $0.0622568 - 1.96(0.0150719) = 0.0327158$, rounds to 0.0327
Upper: $0.0622568 + 1.96(0.0150719) = 0.0917978$, rounds to 0.0918

Answer: 0.0327 to 0.0918

c. (3 points) If the study were expanded to include 900 patients in the drug treatment group instead, would you expect the 95% CI to be (circle one)

i. Broader

ii. Narrower **Yes**

iii. About the same width

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5.

a. (3 points) The following block of R code is intended to compute the standard error of the mean of a vector called `vec`. Please identify and explain any mistakes in the code that would result in an incorrect calculation. If there are none, write "No mistakes".

```
vec <- c(5, 7, 102, 98.6, 8, 85, 64, 54)
sd(vec) / sqrt(length(vec))
```

no mistakes

b. (4 points) An automated hydrophone has been set up at a site to record the number of orca "call bursts" in 10-minute listening windows. Scientists in a coastal ecology lab want to test whether the counts follow a Poisson distribution, using $\alpha = 0.05$. They are unsure what the true mean rate is and will estimate it from the data. They collected the following observed frequencies of call bursts per 10-minute window:

```
> obs
 0  1  2  3  4 5+
18 50 46 20  9  7
```

This yields a mean rate of 1.82 call bursts per 10-minute window. The expected frequencies under a Poisson distribution with estimated mean 1.82 are determined and saved in an object called `exp` [**Note**: you may assume that these have been computed correctly]:

```
> exp
        0         1         2         3         4        5+
24.794833 44.630700 40.167630 24.100578 10.845260  5.460999
```

A hypothesis test is then performed, with output:

```
> chisq.test(x = obs, p = exp, rescale.p = TRUE)

        Chi-squared test for given probabilities

data:  obs
X-squared = 4.8003, df = 5, p-value = 0.4407
```

All code ran without any errors [**Note**: R safely ignored the labels of "0" through "5+" in `obs` and `exp`]. The researcher then writes, "Because $\chi^2 = 4.80$ with 5 degrees of freedom, we fail to reject the null hypothesis (P-value = 0.44 greater than α of 0.05). The patterns of call bursts are consistent with expectations under a Poisson distribution."

Please identify and explain any mistakes in the analysis or conclusions. If there are none, write "No mistakes".

The hypothesis test, as performed, uses the **wrong degrees of freedom** since the mean rate (1.82) was estimated from the data. Because chi-squared = 4.80 with 4 df, we still fail to reject the null hypothesis (P will still be greater than 0.05), so the conclusion itself doesn't materially change.

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6. A polymerase chain reaction (PCR) assay is used to detect a rare allele associated with pesticide resistance in an insect population. Each assay produces either a positive result (allele detected) or a negative result (allele not detected). Among insects that truly carry the resistance allele, the assay gives a positive result 88% of the time and a negative result 12% of the time. Among insects that truly do not carry the resistance allele, the assay gives a negative result 94% of the time. Independent population sampling indicates that 7% of insects in this population carry the resistance allele.

   a. (2 points) For a single PCR assay, the possible outcomes are either a positive result or a negative result. These two outcomes are ____. Circle all options that are correct.
      i. Events with probabilities summing to 1 **Yes**
      ii. Mutually inclusive
      iii. Mutually exclusive **Yes**
      iv. Overlapping events

   b. (6 points) What is the overall probability that a PCR assay result is *correct*? Round your answer to four decimal places.

      = Pr[carrier]*Pr[positive | carrier] + Pr[non-carrier]*Pr[negative | non-carrier]
      = 0.07*0.88 + 0.93*0.94 = 0.9358

      Answer: 0.9358

   c. (4 points) An insect is randomly sampled from the population and its PCR assay is positive. What is the probability that this insect truly carries the resistance allele? Round your answer to four decimal places.

      = Pr[carrier | positive]
      = (0.07*0.88) / (0.07*0.88 + 0.93*0.06) = 0.5247

      Answer: 0.5247

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7. A conservation physiology study investigated whether two different temporary marking methods affected the risk of skin irritation in juvenile sea turtles during a 6-week monitoring period. Turtles were randomly assigned to receive either a non-toxic paint mark on the shell or a small adhesive tag on the carapace. At the end of 6 weeks, researchers recorded whether each turtle developed moderate-to-severe skin irritation near the marking site to address the question: Does the risk of moderate-to-severe skin irritation differ between turtles fitted with each marking method?

Group	Moderate-to-severe irritation	No moderate-to-severe irritation
Paint mark	8	92
Adhesive tag	14	153

a. (2 points) State the null and alternative hypotheses.

H0: Irritation status is independent of marking method
HA: Irritation status is not independent of marking method
(exact wording may vary)

b. (6 points) Fill the table below with expected values and totals, based on H_0. Round values to 4 decimal places.

Group	Moderate-to-severe irritation	No moderate-to-severe irritation	Total
Paint mark	8.2397	91.7603	100
Adhesive tag	13.7603	153.2397	167
Total	22	245	267

c. (3 points) Name a statistical test that you could use to perform the analysis. You do not need to perform the test itself.

Chi-squared contingency test

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8.

a. As you **<u>increase</u>** the size of your sample from a given population, what will generally happen to the following? *For each of the blanks, please write one of:* ***increase, no change****, or* ***decrease****.* (2 points each)

   i. Sample mean: _____no change_____

   ii. Sample standard deviation: _____ no change _____

   iii. Width of the 99% CI: _____decrease_____

   iv. Probability of Type I error: _____ no change _____

   v. Probability of Type II error: _____decrease_____

   vi. Precision of an estimate: _____increase_____

b. (3 points) Is a P-value a conditional probability? State "yes" or "no" and justify your answer by describing what a P-value is in 1-3 sentences.

   Yes, it is a conditional probability. A P-value is the probability of observing a test statistic equal to or more extreme as the one observed, given that the null hypothesis is true. It is conditional on the null hypothesis being true.

c. (3 points) Among the following, circle *all* that are true.

   i. In a normal distribution, approximately 68% of observations fall within 1 standard deviation of the median. **Yes**
   ii. The normal distribution is a discrete numerical probability distribution and the area under the curve equals 1.
   iii. All standard normal distributions have a mean of 0. **Yes**
   iv. The Central Limit Theorem states that the larger the sample size, the closer an estimate is likely to be near its true population parameter value, regardless of the population's distribution.
   v. It is impossible for values of any data that are drawn from a normal distribution to be negative.
   vi. If a random sample of data are normally distributed, the population from which the sample was drawn must be normally distributed.